blood cells (which fight against infection) increases slightly during pregnancy. Blood plasma volume increases to the extent of 1.25 litres (Non pregnant plasma volume is 2500 ml, increase about 3750ml during pregnancy.

Urinary System

- Glomerular filtration rate (GFR) is increased by 50% all throughout pregnancy. Frequency of micturation is a common symptom of early pregnancy. This is due to changes in pelvic anatomy. A degree of hydronephrosis and hydroureter exist. These result from the loss of smooth muscle tone due to the progesterone, aggravated by mechanical pressure from the ureters at the pelvic brim.
- Vesicoureteric reflux is also increased. These changes predispose to urinary tract infection.
- Glycosuria of mild degree in 35% to 50% of all pregnant woman. Increased glomerular filtration leads to more sugar reaching the tubules that can be reabsorbed.

Gastrointestinal System

- The gums become congested and spongy and may bleed to touch.
- Relaxation of lower oesophageal sphincter produces regurgitation and heart burn.
- Slight reduction in gastric secretion and diminished gastric motility result in slow emptying and more effective pulping of food and causes nausea.
- Reduced motility of large intestine increased time for water reabsorption which leads to induce constipation.

Nervous System

- There may be generalised neuritis probably due to Vitamin B_1 (Thiamine) deficiency.
- Compression of the median nerve underneath the carpal ligament over the wrist joint, leading to pain in the hands and arm called **CARPAL TUNNEL SYNDROME** may appear in late pregnancy.
- Nausea, vomiting, mental irritability and sleeplessness due to some psychological background.

7.3 Diagnosis of Pregnancy

First Trimester

Presumptive signs:

- Amenorrhoea-Absence of menstruation.
- Morning sickness- Nausea, vomiting on rising from bed, loss of appetite.
- Frequency of micturition due to congestion of the bladder mucosa.
- Breast discomfort- feeling of fullness and 'Pricking sensation' is present.
- Darkening of the nipples, primary and secondary areolar change.
- Fatigue or tiredness.

Probable signs

- Breast changes: The breasts are enlarged, evident between 6 to 8 weeks.
- Vaginal Sign- The walls become softened and looks bluish in colour. Copious non irritating mucoid discharge appears at 6th week.
- Osiander's sign- There is increased pulsation, felt through the laternal fornices at 8th week.
- Jacquemier's or Chadwick's sign - a bluish discoloration of the cervix, vagina and labia. This is due to local vascular congestion.
- Hegar's Sign- Upper part of the uterus is enlarged by the growing fetus, and lower part of the body of the uterus is empty and extremely soft.

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